

OpenCal

The World's First Open-Source Layerless 3D Printer

Final Capstone Report

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Acronyms & Abbreviations

CAL	Computed Axial Lithography
AM	Additive Manufacturing
3D	Three Dimensional
OSH	Open-Source Hardware
DFMA	Design for Manufacturing and Assembly
FDM	Fused Deposition Modeling
VAM	Volumetric Additive Manufacturing
SLA	Stereolithography
PETG	Polyethylene Terephthalate Glycol
CAD	Computer-Aided Design
LED	Light Emitting Diode
BOM	Bill of Materials
STL	Stereolithography Language
LCD	Liquid-Crystal Display
RTD	Read the Docs
AC/DC	Alternating Current / Direct Current
TPU	Thermoplastic Polyurethane
OTS	Off-the-Shelf

Legend

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Executive Summary [Author: WM; Editor: ML]

Computed Axial Lithography (CAL) is a ground-breaking 3D printing technology that produces parts volumetrically in a single step, rather than using a repeated layering process like in traditional additive manufacturing (AM) [1]. This layerless technique offers numerous advantages over traditional AM methods, such as faster print times of complex geometries without the need for supports. In its most basic form, CAL consists of a rotating vial of photopolymerizable resin and a light source that displays a projected video. As the vial rotates, the resin is selectively cured to form a part. Despite its advantages, CAL has remained largely confined to research laboratories due to complex optical setups, high equipment costs, and limited access to materials, limiting its potential applications in education, hobbyist manufacturing, and makerspaces.

Historically, open-source hardware (OSH) has played a transformative role in broadening access to AM. For CAL to reach the same level of democratized innovation seen in traditional 3D printing, it needs to overcome barriers to manufacturability, accessibility, and usability. OpenCAL, the world's first open-source, layerless 3D printer, developed at UC Berkeley by Waddell *et al.* in 2025, aimed to do just that [4]. This second iteration builds upon that foundation by redesigning its hardware, electronics, software, and post-processing systems using principles of design for manufacturing and assembly (DFMA). Through accessible documentation and an online community, several pilot studies are underway to build OpenCAL, accelerating innovation in CAL technologies. Future work will involve a public release and continued refinement based on feedback from these pilots.

I. Introduction [Author: WM; Editor: ML]

Computed Axial Lithography (CAL) is a novel 3D printing technology that forms parts volumetrically in a single step, rather than using a repeated layering process as found in traditional additive manufacturing (AM) [1]. This technique offers numerous advantages over traditional AM methods, such as faster print times of complex geometries without the need for supports. The basic components of the technology consist of a vial of photopolymerizable resin and a light source that displays a projected video. The part is formed as the vial rotates about its Z-axis at the same speed as the projected video, selectively curing the resin (Figure 1).

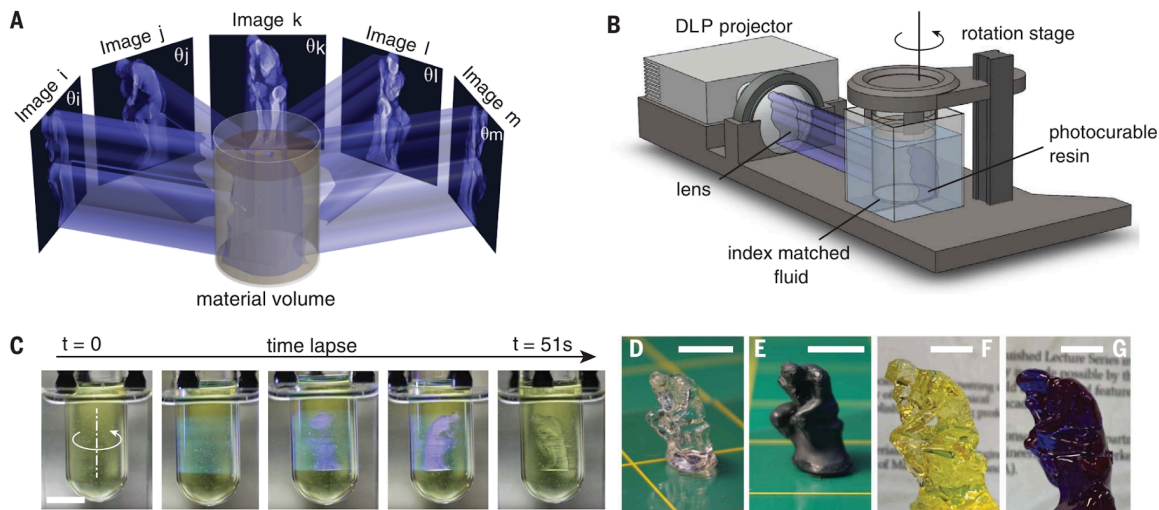


Figure 1. CAL Fabrication Process [1]. (A) Underlying mechanism. (B) CAL system schematic. (C) Part printing process. (D-G) Example *Thinker* statue prints in varied finishes.

Despite its advantages, CAL has remained largely confined to research laboratories since its inception about a decade ago, due to complex optical setups, high equipment costs, and limited access to materials. As a result, its development has been driven by a small academic community, slowing broader experimentation and limiting its potential applications in education, hobbyist manufacturing, and makerspaces.

Historically, open-source hardware (OSH) has played a transformative role in broadening access to AM. Projects like RepRap catalyzed the rise of commercial layered 3D printers, by enabling the public access to modify and improve shared designs, thus creating reinforcing cycles of innovation [2, 3]. For CAL to reach the same level of democratized innovation seen in traditional 3D printing, it needs to overcome barriers to manufacturability, accessibility, and usability. OpenCAL, developed at UC Berkeley by Waddell *et al.* in 2025 as the world's first open-source, layerless 3D printer, aimed to do just that [4]. This project builds upon that foundation by redesigning its hardware, electronics, software, and post-processing systems using principles of design for manufacturing and assembly (DFMA). This version of OpenCAL is accompanied by thorough and accessible open-source documentation. We hypothesize that reducing complexity and improving usability will lower barriers to adoption and accelerate innovation in layerless 3D printing.

II. Current Methods and Technologies [Author: ML; Editor: WM]

AM is widely implemented across industries as a means of producing complex components without tooling and with reduced material waste compared to subtractive manufacturing processes [5]. Popular commercially available systems are often either based on material extrusion, such as fused deposition modeling (FDM), or vat polymerization, such as stereolithography (SLA). FDM works by depositing extruded thermoplastic material to form parts, while SLA printing uses a focused laser beam to selectively cure resin [5]. By relying on depositing material layer-by-layer, these techniques are limited by factors such as part geometry and layer resolution.

Volumetric additive manufacturing (VAM) is an emerging alternative to traditional layered AM in which the entire volume of a part is produced in a single step through

tomographic light projection and photopolymerization [1]. This approach eliminates many of the limitations of layered printing and provides novel applications for 3D printing. Without layers, VAM can provide higher resolutions, essential for applications where precision or a smooth finish are important. Since components are formed without supports, objects can be placed within the resin vial during printing for overprinting and the formation of geometries around existing components. Printing for multi-material parts can be applied to encapsulating electronics, dentistry, soft robotics, and more [1].

Commercial implementations of VAM remain limited and are primarily focused on specialized applications. Biomedical printing companies such as Readily3D utilize layerless printing with photopolymerization to create high-resolution biocompatible components [6]. Printers like these rely on proprietary hardware and are prohibitively expensive for certain applications. This is because the tomographic projections must be extremely precise and account for factors such as light attenuation, fluid motion, refraction, etc. This means hardware, software, and optical components must be finely tuned and typically expensive. Additionally, the material properties of the printing resin must be carefully controlled [7]. OpenCAL aims to provide a more affordable, open-source alternative to this technology, balancing high print quality and reproducibility while lowering part cost and expertise needed for assembly and operation of the device.

III. OpenCAL Approach [Author: BV & ZS; Editor: PM]

As the majority of CAL technology is under patent, such as Patent WO2018208378A2, OpenCAL will revolutionize the industry by providing all the necessary methodologies for replicating the fundamentals of a CAL system [8]. The entire system is designed in Onshape, a free and collaborative CAD software. The documentation for OpenCAL assembly and operation

will be hosted on ReadtheDocs, a platform chosen for its maintainability and public availability. All necessary software will be available on a public GitHub repository and a Discord server has been established to grow an online community.

The OpenCAL device is broken down into 6 main subsystems: Frame, Optics, Electronics, Rotation, Post-processing, and Software. The frame is made entirely of 20 mm x 20 mm aluminum extrusions due to their commercial availability, low cost, strength, and versatility. For similar reasons as the frame and to simplify manufacturing, the enclosure is made from black acrylic. All 3D printed components were designed to be able to fit on a Prusa Core One+ build plate (250 x 220 x 270 mm), a hobbyist printer. All printed parts were also printed in PETG for its durability and chemical resistance.

The optics consist of a standard projector, a Fresnel Lens, and a 3D-printed shroud. The device works by projecting a rotating 3D image of the desired part via any commonly available projector. A NexiGo Nova Mini Laser Projector was selected for its commercial availability, broad focal range, and compact size. However, the optical system is parametrically designed to be modified for a wide range of household projectors, requiring only their dimensions, lens positions, and throw ratios. The projector mount interfaces with the aluminum railing and allows fine-tuning of the alignment. A Fresnel lens collimates the light from the projector onto the vial, and a 3D printed shroud minimizes reflections and other external light from interfering with the optics. The rotational element subsystem includes a stepper motor, a rotational stage with bearings to align and rotate the vial during the print. A Raspberry Pi Camera HD v3 12MPx is mounted behind the rotation element to enable automatic print completion detection [9, 10]. The projections are calculated such that all resin within the part volume solidifies simultaneously, leaving everything else liquid.

After the part cures, the vial is removed and placed in a centrifugal chamber (Centrifugal) where excess resin is forced out. The post-processing subsystem is designed to remove uncured resin from the vial and isolate the printed part. The vial is spun at the optimal angular velocity, driving excess resin outward and significantly improving cleaning efficiency within the vial. Following similar design principles as the rest of OpenCAL, this subsystem is largely composed of 3D printed PETG components.

Many components in the electronics system are standard for existing 3D printer technologies, such as the Nema stepper motor and the TMC2209 driver, providing both noise reduction and power. The electrical distribution consists of a single outlet connection to a 24V AC/DC adapter, with several buck converters for all component voltage ratings. An LCD screen serves as the main user interface for OpenCAL. Some electronics are present, not for base functionality, but for extending the application of this technology in the future, an important facet of open-source design

The software is divided into two systems: VAMToolbox, which runs on any computer, and the OpenCAL software, which runs locally on the Raspberry Pi that controls the device. VAMToolbox allows users to generate the proper projection videos based on any 3D part's STL file. Since it is computationally intensive, it is distributed as a separate computer application. In contrast, the OpenCAL software is lightweight and simply controls the user interface, stepper motor, projector, LEDs, and camera on the device. The software is designed to be extendable by the community, and focuses on precise calibration and control of the electrical components to enable accurate prints.

Thorough documentation will be necessary for good open-source access. It will contain a Bill of Materials (BOM), CAD files, and detailed step-by-step assembly instructions. After

completing the first OpenCAL prototype in early March 2026, the project will undergo several pilot programs with other academic institutions and industrial partners to streamline the design and documentation before the initial public release in May.

IV. Results

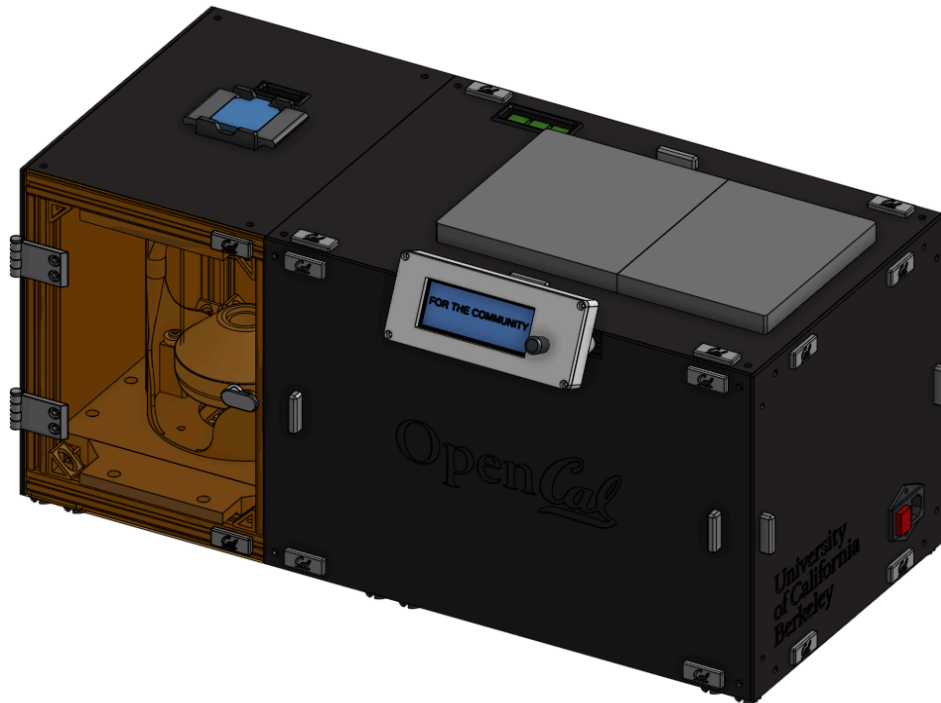


Figure 2. Isometric View of OpenCAL

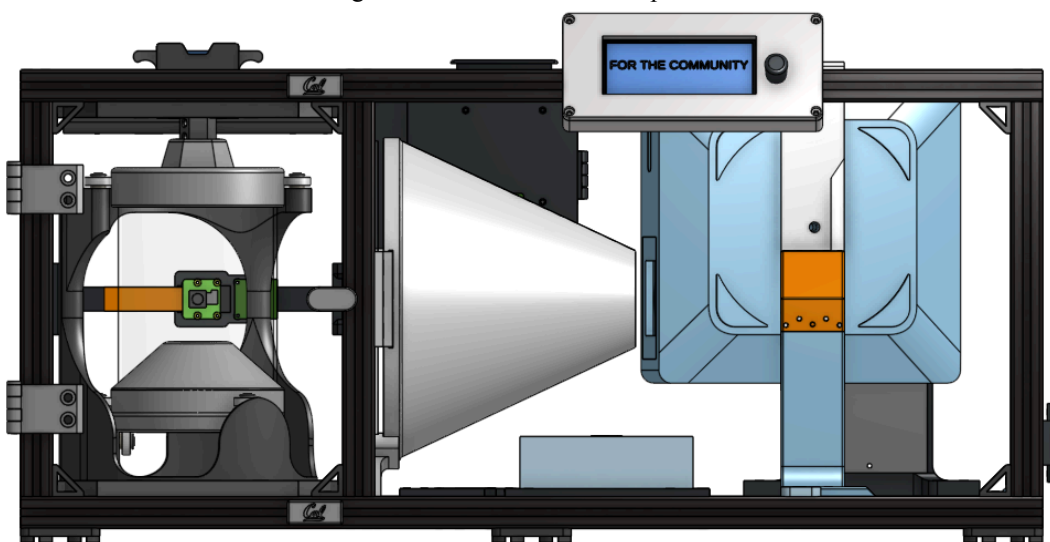


Figure 3. Front View of OpenCAL without Paneling

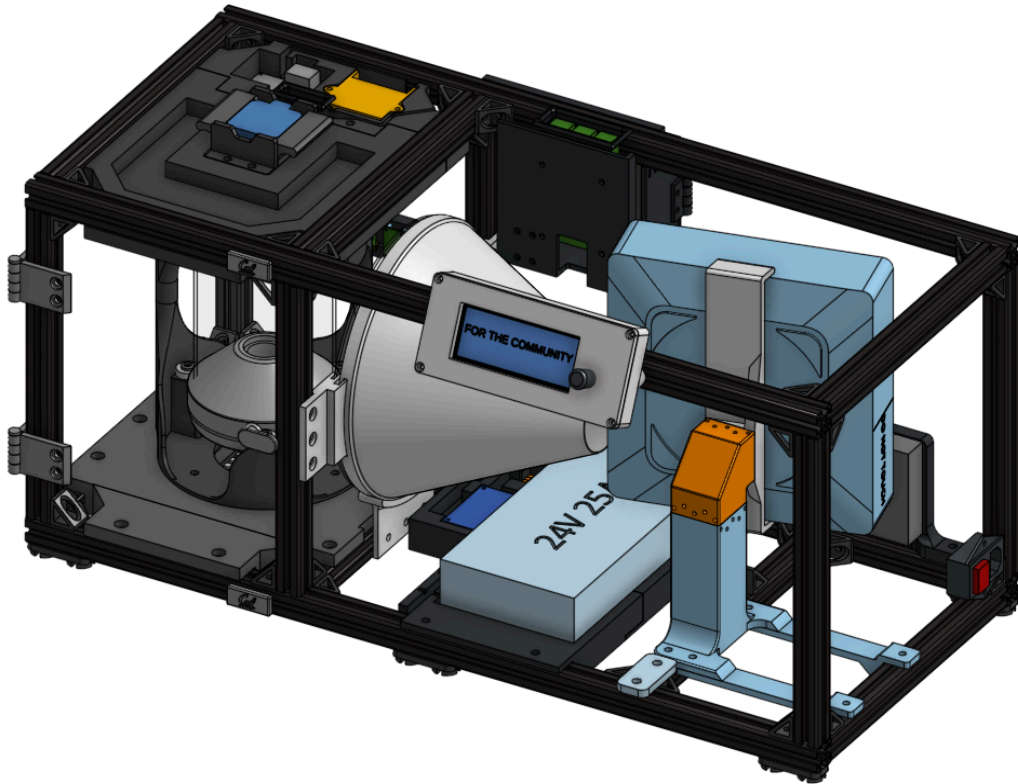


Figure 4. Isometric View of OpenCAL without Paneling

Technical Specifications:

- Cost: ~ \$1,717.66 USD
- Weight: ~ 12 kilograms
- Maximum Volume Dimensions (LxWxH): 670.5 x 371.2 x 334.81 mm
- Runtime for Small Vial Print: ~ 1.5 minutes
- Vial Axial Deviation: < 0.3mm

Bill of Materials (BOM):

- [BOM Link](#)

The rest of this section will go through, in detail, each of our design decisions and how we are implementing various subcomponents into the overall system.

i. Frame and Paneling [Author: BV; Editor: ML]

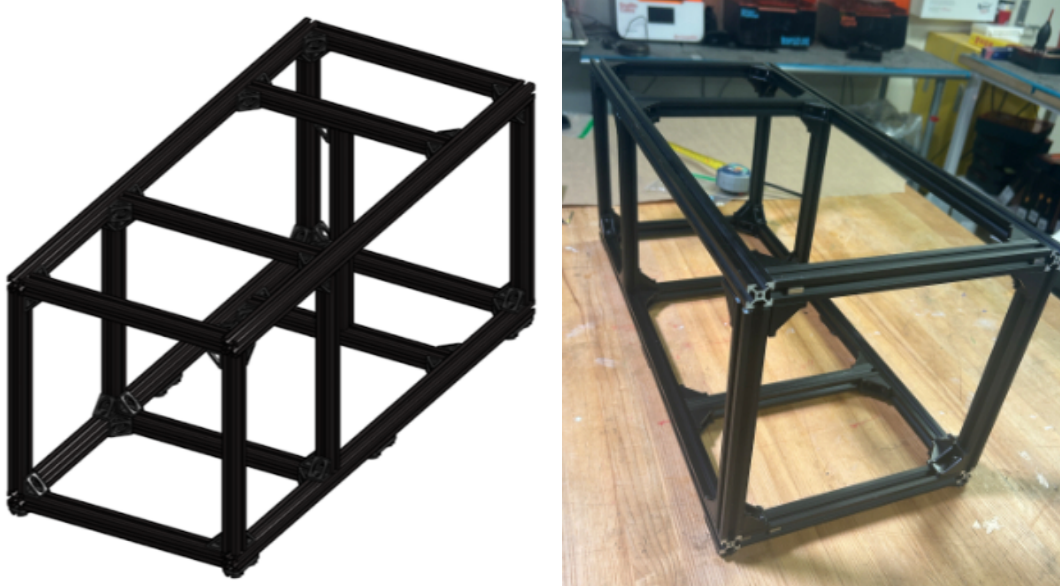


Figure 5. Frame CAD (left) and build (right)

The frame, seen in Figure 5, serves as the main structure of the printer, where every other system is installed. It is built entirely of 20 x 20 mm aluminum extrusion with only two different lengths: 250 mm and 650 mm. These specific lengths were selected such that the printer dimensions allow for a wide range of different home projectors to be mounted onto the system. Simultaneously, it was designed to perfectly fit a Pelican 1630 case, as the printer will undergo transport across the country for conferences and events. Underneath the frame are six 3D-printed TPU feet, which are designed to absorb and minimize vibration from both the printer and the environment. The frame also includes 12 magnet frame inserts, 3D printed magnet holders that have 12x5x2 mm magnet bars super-glued within them. These slot into the railing and interface with the magnets on the paneling.

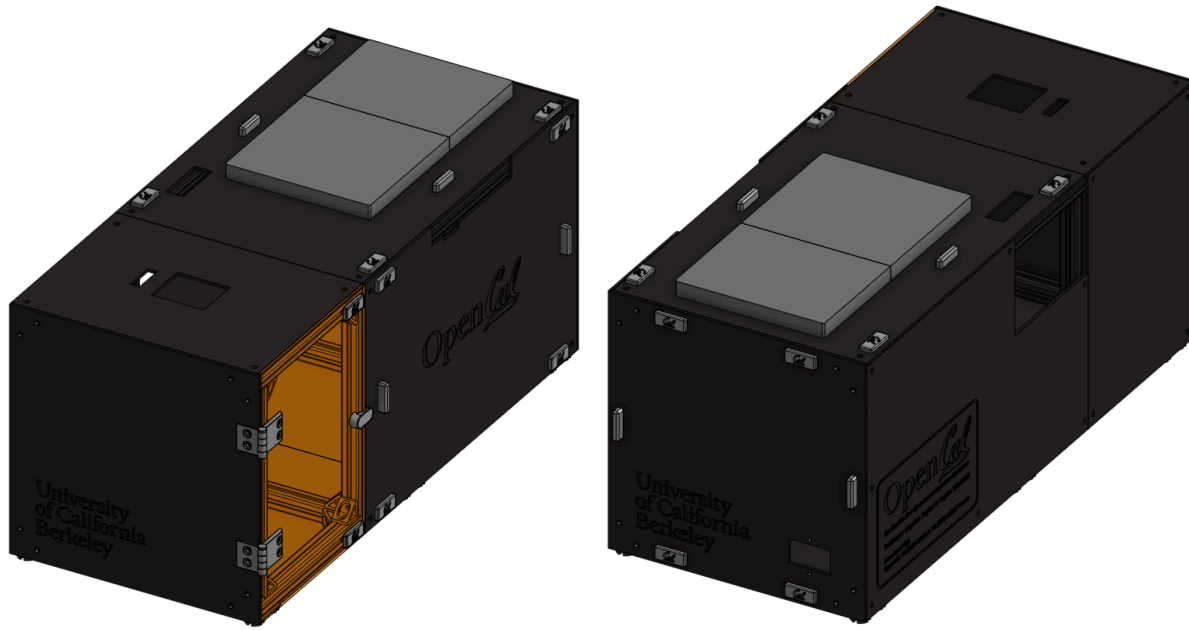


Figure 6. Isometric Views of the Front (left) and Back (right) Paneling CAD

The panels, shown in Figure 6, serve to block out ambient light (specifically blue light) that can interfere with the photosensitive resin, while also providing protection and aesthetics for the printer. Each panel is a black laser-cut 3mm acrylic, with the door being a transparent orange 3mm panel to provide print visibility, while still blocking out blue light.

Three of the panels (those interfacing with the projector) are magnetic for maintenance purposes. These panels each have 4 3D printed magnet holders that each have two of the 12x5x2 mm magnet bars super-glued into it. Two magnets are used in this case to induce a stronger magnetic field while fitting in the rail to physically lock onto the Magnet Frame Inserts mentioned previously. The magnetic panels also feature two 3D printed finger holds. The door features a 3D printed handle and hinges, and magnetically adheres to the frame when closed. Each of the non-magnetic panels has multiple M5x8 screws and corresponding M5 T-nuts so that the panels can slot into the railing and be locked down. The top panel features a 3D printed projector extension that can be customized to accommodate larger projectors. As the projector

may actually exceed the dimensions of the printer volume, protruding from the top, this was added to further increase the range of projectors compatible.

ii. Optics [Author: BV; Editor: WM]

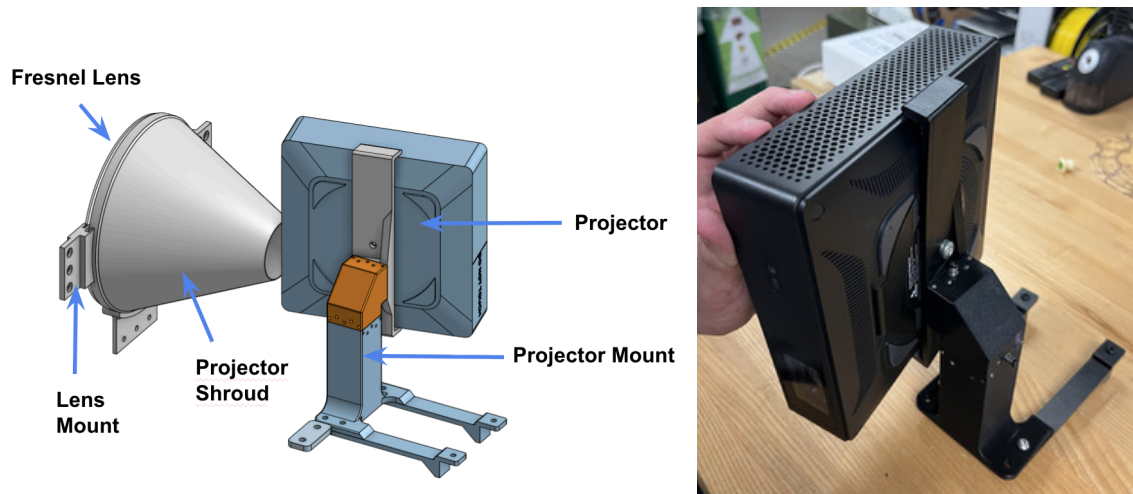


Figure 7. Optics CAD systems (left) and built a projector mount (right).

The Optics subsystem provides the energy and the image to the resin. It features the NexiGo Nova Mini Laser Projector, which is around 197x197x60 mm. Most projectors that are this size or smaller should work with our system. One recommendation is for the projector to have a good focus and keystone ability, a throw ratio of 1.2:1, and a “boost mode” to allow for higher quality. However, it is required that the projector has a center mounting hole to mount with our custom mount. The projector mount shown in Figure 7 is a 3D printed XYZ stage, which allows the projector to be moved in each of the X, Y, and Z directions by an additional ~ 10 mm. This accounts for any tolerancing issues that may occur during assembly. It uses springs, 3x40mm steel rods, nuts, and screws to function. Much of the calibration for the projector mount is done via CAD. Rather than having to move the projector around to find the optimal spot for collimation, both X and Y locators attach to the railing, and the projector mounts to position it on

the frame. The lengths for these locators are based on equations in Onshape that users may edit based on their specific projector.

The Optics subsystem also involves a 200mm Fresnel Lens, Lens Mounts, and a 3D printed Projector Shroud. Through collimating the light rays from the projector, all the rays become parallel, which allows the image to be the same size across the entire vial, which is critical for printing (see Figure 6). The lens is held in place using 3D printed mounts, which also serve as the mounts for the Projector Shroud.

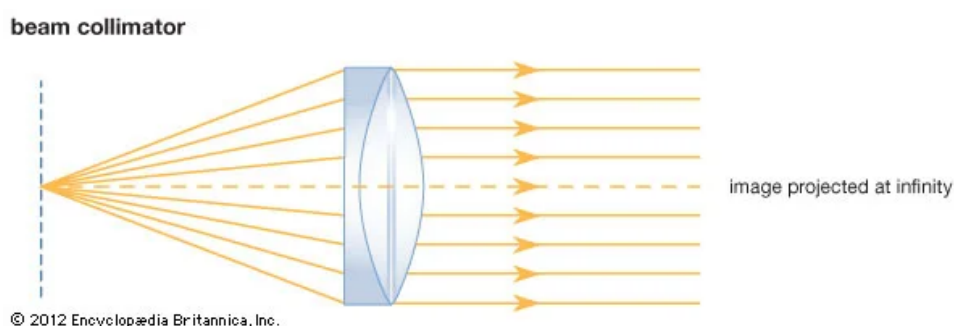


Figure 8. How Light Collimation Works. "Collimator," Encyclopædia Britannica, Encyclopædia Britannica, Inc.

[Online]. Available: <https://www.britannica.com/science/collimator>

iii. Rotational Element [Author: WM; Editor: ZS]

The Rotational Element subsystem supports the Z-axis alignment and smooth rotation of the vial of resin. The rotational stage can accommodate a maximum glass vial height of 6in (152.4mm) and diameter of 4in (101.6mm), and is shown as the large vial in Figure 9 (left). Smaller vial sizes, such as those depicted in Figure 9 (right) can also be accommodated through parametric vial holders, which help center the vial in the rotational stage.

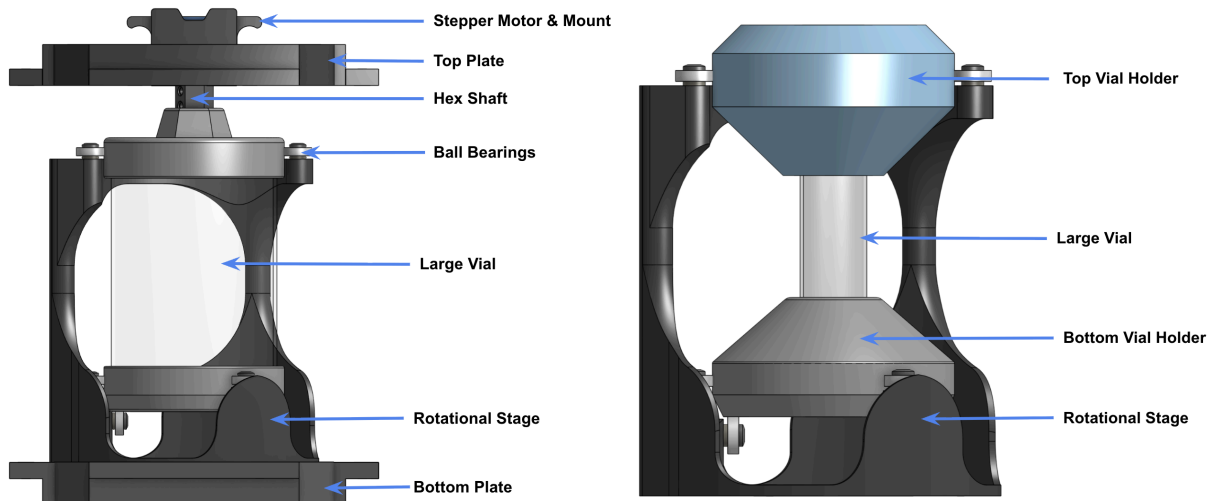


Figure 9. Rotational Element Subsystem with Large Vial (left) and Small Vial (right)

The large vial is designed to be open-ended to facilitate ease of cleaning out the resin. Attached to each end of the large vial is a 3D printed glass attachment that locks with the lid to compress an oring to create a watertight seal. The top lid features a hex key for the hex shaft of the stepper motor to interface with, while the bottom lid features a circular cutout where a 3in (76.2mm) glass disk is attached to the 3D print. This feature enables red light from the LED housed below the bottom plate to illuminate the part during printing. The top and bottom lids are designed to twist off in opposing directions, such that removal of one end does not accidentally lead to removing the other end.

To fit typical hobbyist 3D printer build plates, the top and bottom plates were printed in parts. The top plate is divided into two parts: a front plate, which features the stepper motor and stepper motor mounting plates, and a back plate, which houses some of the electrical components. The stepper motor is mounted using a plate with two levels of magnets to facilitate ease of engaging and disengaging the motor from the vial. The bottom plate is divided into three parts: a front base plate, a mid base plate, and a back base plate, allowing the center piece holding the rotational stage to be printed in one part.

iv. Electronics [Author: ML; Editor: ZS]

The electrical system provides power to the various electrical components throughout OpenCAL. It consists of electrical parts and their 3D printed housing. User input is prompted from an LCD display mounted at the top of the front of the printer, controlled with a rotary encoder knob. This resembles the UI design of existing commercial 3D printers. The LCD and other electronic components are visible in Figure 10.

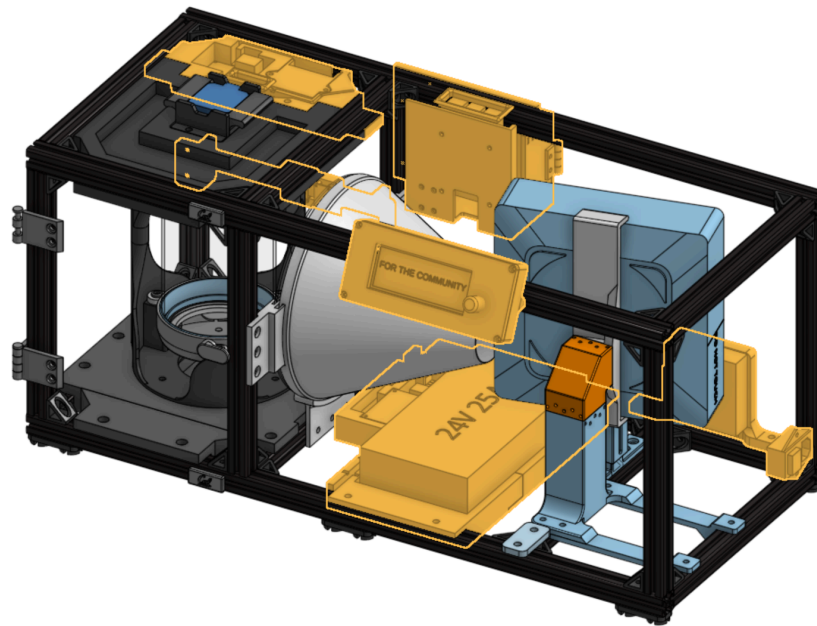


Figure 10. Electronic Component Locations

The system operates on a Raspberry Pi 5 microprocessor, providing an open-source and development-friendly means of control. Requiring the most user interaction of any electrical module, the Raspberry Pi housing includes a hinged door for ease of access, shown in Figure 11. A fan placed on the door diverts heat outside through a set of vents for cooling. Print files are uploaded to the Raspberry Pi through USB ports, accessible through an opening at the top of the printer.

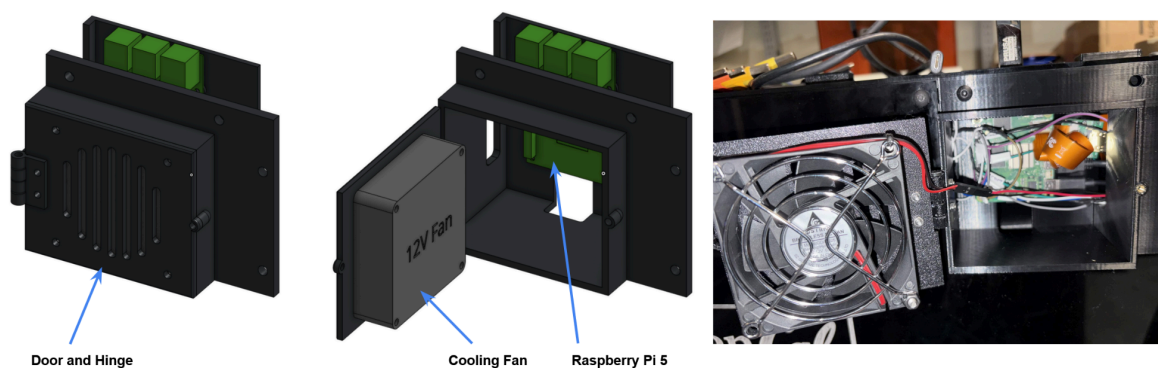


Figure 11. Raspberry Pi 5 Housing

The motor has to be capable of supplying enough torque to smoothly rotate vials of various masses. The motor selected for this application, a Nema 17 stepper motor, provides a combination of torque, up to 56 Ncm, and precision, operating at 1600 microsteps per revolution. At the small scale OpenCAL operates, motor precision is pertinent to print resolution. A TMC 2209 driver, a common choice in 3D printing applications, was selected for torque and quiet performance.

Additional components increase the functionality of the printer. A set of RGB LEDs, a square array of Neopixel lights, integrated beneath the rotation stage, provides a simple and compact way to shine light through the bottom surface of the vial. The array emits red light, outside of the spectrum of light wavelengths capable of triggering a photochemical reaction in the resin, illuminating solid component outlines within to make determining print completion easier.

Several components run off different voltages. To accommodate this while utilizing a single outlet, an adapter converts 120V AC power from the outlet to 24V DC. Three buck converters drop the voltage supplied by the converter to 5V, 12V, and 19V, providing power based on the specifications of each component. Made to be modular and customizable, removable plugs such as screw terminals and DuPont connectors were favored over permanently

soldered connections. The wiring diagram in Figure 12 shows the connections between components.

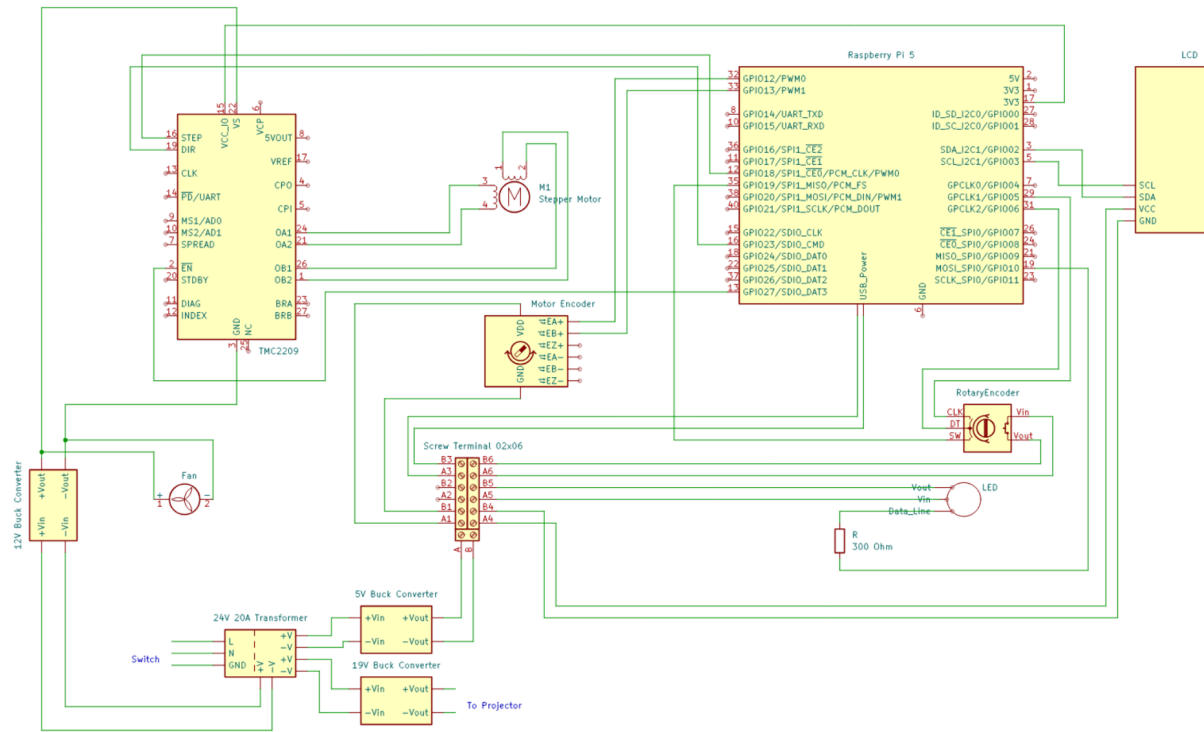


Figure 12. Wiring Diagram

As wires obscuring the image from the projector would be detrimental to print results, different tools for cable management were implemented. Cable clips, designed to sit snugly in the 80-20 frame, keep wires constrained. Press-fit cable clips in 3D printed components also keep wires within designated wire routing paths. Openings for zip ties help keep larger bundles of wire in place.

v. Post-Processing and Centrifugal [Author: ZS; Editor: ML]

The Post-processing system is extremely important for OpenCAL, as it plays a major role in the final quality of parts and general user safety. The use of photopolymerizable resin has health risks, such as skin and lung damage, if not properly handled. To mitigate this risk, the

post-processing workflow aims to contain uncured resin and ensure safe use of the OpenCAL system without worrying about the risks of handling hazardous materials. The post-processing workflow is split into two processes to ensure optimal results, one for vial emptying and one for curing. The curing process was fully developed before the work done by this team, so this section will only focus on the vial cleaning process, known as Centrifugal.

Part extraction after printing with OpenCAL is often one of the most frustrating steps for users. The printed part's natural translucency makes it difficult to locate when submerged in resin, increasing the likelihood of mishandling during removal. To address this, the centrifugal cleaner provides a safer, more controlled extraction method, eliminating the need for manual retrieval from the vial.

The device operates by spinning at high speeds using a planetary gearbox with an approximate 3:1 ratio, enabling efficient torque transmission with minimal losses. It is able to reach rotational speeds of 300-500 rpm, and aims to achieve optimal rotational speeds of around 1000 rpm (as measured, though not formally published) in a future project with an added motor. At these speeds, the system not only facilitates extraction but also effectively removes partially cured resin adhering to the part. This dual functionality significantly streamlines post-processing, reducing the intensity of cleaning typically required in other resin-based techniques such as stereolithography (SLA).

The centrifugal design was made almost entirely with 3D printed parts, ensuring the device is readily accessible as a cheap add-on to the OpenCAL system. Optimizing for ease of fabrication, minimal supports are required. The final design can be found in Figure 13.

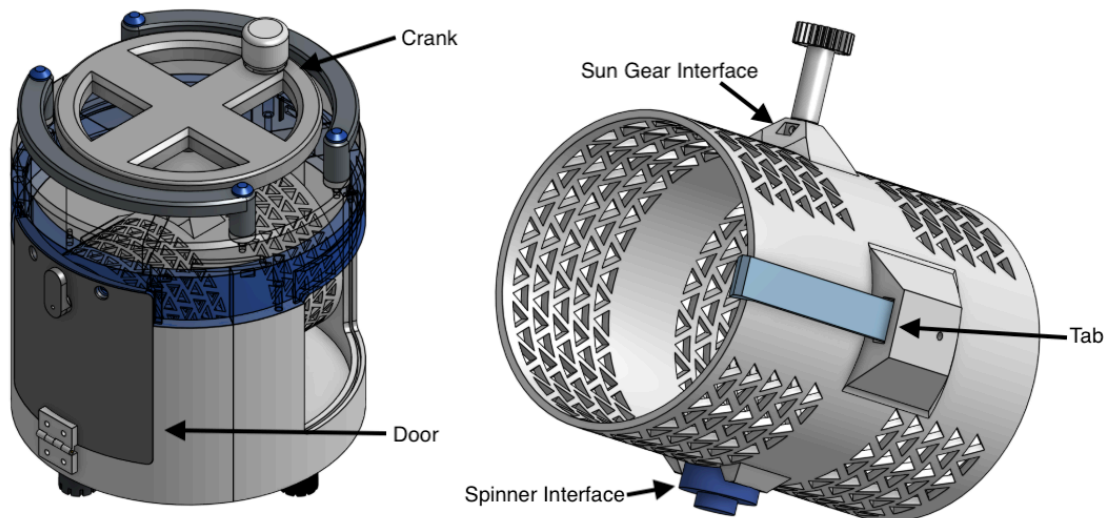


Figure 13. Finalized Centrifugal Design (left) & Vial Holder Subassembly (right).

A door loading mechanism was decided on to enable easy loading and unloading of the vial. The door is held closed by magnets and has ridges on the inside to keep any resin from escaping.

The vial holder is designed to hold the vial securely while spinning, have easily interfacing components, and have visibility for users to check on progress. A tab was added to one side to hold the vial in place, and it can be easily removed or replaced, and the opposite side has ridges to keep the vial from slipping. Triangles have been added to allow for visibility without requiring supports for printing. The top has an interface to the sun gear that, when secured, connects the system to the crank, allowing it to spin. The bottom of the vial holder interfaces with the spinning component, which sits on a metal bearing to ensure smooth rotation. The subassembly of the vial holder can be seen in Figure 13.

One of the other major design challenges was with the lower part of the base (Figure 14). To accommodate a smaller print bed size, the base was divided into several pieces. The base slopes downward, guiding resin to drain out the bottom.

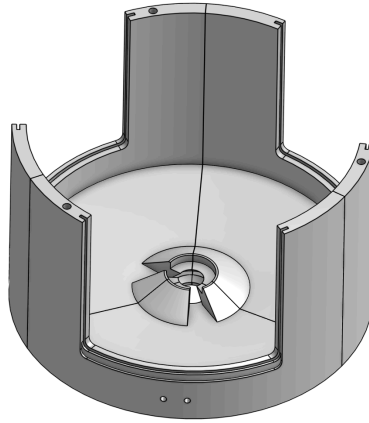


Figure 14. Lower Base

Final testing has shown that the Centrifugal system performs as expected, and it is a functional way to remove the excess resin from the part.

vi. Software [Author: PM; Editor: BV]

The OpenCAL software is separated into two components: VAMToolbox converts part geometry into projection videos (MP4), while the OpenCAL library controls the device. Both software components are developed and installed separately. This design choice improves modularity and minimizes the computational needs of the printer.

VAMToolbox exists independently of OpenCAL and provides the general mathematical methods for calculating optical projections based on the target part geometry. It is entirely written in the Python language and leverages established computational libraries like NumPy and PyTorch. It also accounts for projector and material properties, which affect how light will interact with the system. VAMToolbox was in development before the OpenCAL project, and the majority of the mathematics was completed earlier. However, as part of the OpenCAL open-source initiative, VAMToolbox was adjusted and reorganized to improve usability. Software dependencies were simplified to streamline the installation process. Additionally, basic VAMToolbox usage that would be most applicable to OpenCAL users was packaged into a more

application-like workflow. Users can install VAMToolbox on any laptop/desktop computer and use it to convert STL files into MP4 videos that the OpenCAL printer can use. Since VAMToolbox is computationally intensive, running it on a separate computer greatly reduces the printer's required computational power, therefore also reducing the cost.

Separately, the OpenCAL library controls the physical printer and all connected electronics. It is also written almost entirely in Python, with Bash and C scripts used occasionally where necessary to control system processes. Python was chosen for rapid prototyping and easy future extensibility by the community. The software is very lightweight and can be run on a Raspberry Pi 5 with 2 GB of memory. The software controls the LEDs, motor, projector, camera, and user interface. The user interface involves a basic 20x4 character LCD and a single knob/pushbutton, providing the necessary controls and feedback for the user while minimizing cost. The UI and operation logic were designed with a focus on speed, reliability, and simplicity. The goal is for users to easily augment or add functionality they find useful. The main UI features include: selecting part videos and rotation speeds, toggling the camera and LEDs, and tools for evaluating/calibrating the optical alignment. Examples of some optical alignment tests can be seen in Figure 15.

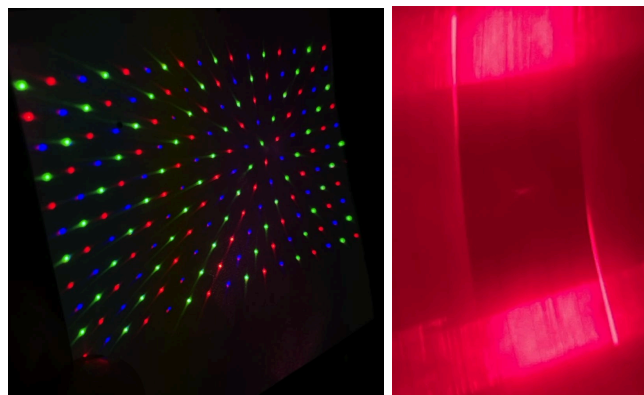


Figure 15. Example projector calibration images. Lens spherical and chromatic aberration (left) and projector vertical angle alignment (right) are being visualized.

vii. Open-Source Instructions [Author: BV; Editor: PM]

The OpenCAL project is documented publicly at the following link: <https://opencal-org.readthedocs.io/en/latest/>. The website is hosted on ReadTheDocs, a public text-based website rendering service, and is updated whenever any changes are made to the physical device or software. The website documents all materials, resources, and steps necessary to recreate and use the OpenCAL system. Each step includes images and a material list. Additionally, STL files for all 3D-printed OpenCAL components can be found on Printables: https://www.printables.com/@OpenCAL_4568832. Finally, there is an OpenCAL Discord server where users can communicate and seek support.

viii. Print Results [Author: PM; Editor: BV]

During design iteration, the team used test models to evaluate print quality. One recurring model was a miniature *The Thinker* sculpture, chosen for its smooth, detailed features, gaps, and overhangs. The very first prints were featureless blobs, indicating poor optical alignment, motor/video timing, and material qualities. The next set of prints visibly resembled *The Thinker*, but were warped and lacked the proper gaps. After improving alignment and projector stability, print quality improved considerably, yielding fairly consistent, recognizable models. Finally, a larger-scale print (Figure 16) confirmed alignment throughout the vial volume and enabled inspection of the model's finer details.

The team hopes to continue print quality analysis, particularly with more standardized and geometric models. The goal is to quantify print resolution and how it varies across the entire vial diameter.



Figure 16. Layerless printed model of *The Thinker*. Print time was approximately 3 minutes.

V. Limitations & Future Work [Author: ML & ZS; Editor: WM & BV]

As the first implementation of an open-source VAM technology, there are some limitations to current printing capabilities. All of the testing and calibration for OpenCAL has used a resin volume within a 30 mm diameter and 70 mm tall vial (small vial). Further testing is required to accommodate printing in the maximum resin volume within a 100mm diameter and 150mm tall vial (large vial). Future work should implement a wider range of vial sizes to more closely accommodate printed part sizes and prevent waste since any excess, uncured resin material in the vial becomes unusable.

The post-processing system also has room for improvement, as it is currently optimized to be inexpensive and convenient rather than providing optimal performance. In the future, a more robust product will be developed to do the whole post-processing workflow (curing and vial cleaning) in one device.

Designed with manufacturing accessibility in mind, the current version of OpenCAL has some hardware limitations. Adjustments to the design and better tools for calibration can help

account for compounding tolerance issues inherent in 3D printed assemblies. In addition to hardware-based alignment, software calibration could offer a way to ensure quality prints. A software approach could offer multiple tools for adjustments, such as video angle offset and tools for affirming optical alignment accuracy.

Designed to be modular and iterative, this version of OpenCAL has electronic components that are not required for the base functionality of a VAM system, but will be a useful tool for builders and researchers to expand on the technology. An example of this is the Raspberry Pi Camera Module 3, which, through further software development, could automatically determine print completion based on the clarity of print outlines and stop the print to prevent excess curing. Similarly, the motor comes with a pre-attached encoder, which can enable future use cases, such as overprinting [1], where tracking motor position would be necessary.

Public outreach broadens access and interest in OpenCAL, making increasing awareness of the technology a priority. This year, OpenCAL was featured at the Society for Biomaterials Annual Meeting and Exposition in Georgia, and at Rapid + TCT in Boston, large conferences in the world of manufacturing technologies. Another approach is leveraging content creators for outreach. This includes a planned collaboration with the content creator 3D Printing Nerd. We will also be presenting OpenCAL this summer at OpenSauce, which hopes to bring in much more awareness and excitement for the project. With this outreach, we hope to invite the community to make incremental improvements to hardware and software in OpenCAL.

As feedback comes in from external teams working on completing this build, changes are actively being made to the system and documentation. Beyond the completion of OpenCAL as

an MEng capstone project, it will exist as an evolving open-source tool for expanding the development of CAL technology.

VI. Conclusion [Author: PM; Editor: WM]

Over the course of the year, the OpenCAL team has successfully designed and built a functional CAL system that is entirely open-source and accessible to the community. Throughout the design process, decisions were made to prioritize ease of manufacturing, assembly, and maintainability. Components are 3D-printed where possible, and individual subsystems can be disassembled, repaired, or modified independently of each other. The device can be easily transported, and is at a publicly affordable price point of ~\$1,700. The team was able to demonstrate layerless prints that form in under five minutes, and will continue to work on improving device alignment and print quality. Most importantly, all resources and instructions required to replicate the OpenCAL system are available publicly online. This includes an itemized bill of materials, detailed stepwise instructions, and source code. Ultimately, OpenCAL is a significant first step toward improving the accessibility and popularity of layerless 3D printing.

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